The Illusion of Intelligence
智能的幻象

Teaching and Learning with Overconfident AI Agents
与过度自信AI智能体共处的教学与学习

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Learning vs. Teaching
学习 vs 教学

Students - 学生:

- understand concepts – 理解概念
- summarize material – 总结学习内容
- solve assignments – 完成作业
- seek support – 寻求帮助
- explore topics – 探索主题

Faculty - 教师:

- highlight the essential – 突出核心内容
- generate examples – 生成示例
- design activities – 设计教学活动
- provide support – 提供支持
- open perspectives – 拓展视角

Timeless pursuits. Evolving resources.
永恒的追求，不断演变的工具。

The shift: from tools to agents
从工具到智能体的转变

Traditional tools - 传统工具:

- writing - 写作
- calculators - 计算器
- IDEs - 集成开发环境
- search engines - 搜索引擎

Tool = Obvious collaborator
工具 = 明显的协作辅助者

AI agents - AI智能体:

- explain - 解释
- suggest - 建议
- critique - 批评与反馈
- generate - 生成
- imitate expertise - 模仿专家能力

Fluency ≠ Understanding
语言流畅 ≠ 真正理解

The danger starts when tools become *socially persuasive*.
当工具开始具备社会说服力时，危险便随之出现。

The illusion of intelligence
智能的幻象

AI agents - AI智能体:

- ✓ sound (very!) confident - 听起来(非常)自信
- ✓ explain convincingly - 解释得很有说服力
- ✓ produce polished answers - 生成精致的答案

- ✗ no understanding - 没有真正理解
- ✗ no consistency - 没有致性
- ✗ no reasoning - 没有推理能力

Performance is ***not*** *competence.*
表现不等于真正能力。

Classroom reality: student behavior
课堂现实:学生行为

Expected - 预期行为

- ask for hints - 寻求提示
- explore conceptual questions - 提出概念性问题
- verify understanding - 验证理解

Observed - 实际观察

- generate complete solutions - 直接生成完整答案
- debug by trial and error, copy and adapt, "vibe code" - 通过试错进行调试, 复制并修改, “氛围编程”

*Cognition is **outsourced**!*
认知正在被外包!

Example: The Dining Philosophers problem
案例：哲学家进餐问题

Goals

- Guarantee safety (no deadlocks)
- Ensure liveness (highest possible concurrency)

Example: The Dining Philosophers problem
案例：哲学家进餐问题

Bad solution

- Every philosopher is a rightie (right-handed)
- They all pick the right chopstick first at the same time
- They're all stuck with a single chopstick

=> Deadlock!

Example: The Dining Philosophers problem
案例：哲学家进餐问题

Most likely solution from AI

- One philosopher is a leftie
- All others are righties

=> Guarantees safety (no deadlock)

But liveness is not optimal (all righties might starve)

Example: The Dining Philosophers problem
案例：哲学家进餐问题

A better solution

- odd philosophers are lefties
- even philosophers are righties

=> Guarantees safety (no deadlock)

=> Liveness is optimized (average case: half of the table is eating)

What AI does not tell students
AI不会主动告诉学生的内容

Unless it is specified in the prompt, AI often omits
除非提示中明确要求，否则AI往往会忽略：

- murky complexity: tradeoffs, best practices - 复杂但关键的问题：权衡、最佳实践
- optimization criteria - 优化指标
- deeper design alternatives - 更深层次的设计替代方案

AI gives canonical answers. Education often needs *deeper reflection*.
AI倾向于给出"标准答案"，而教育往往需要更深入的反思。

Hidden cost: skill atrophy
隐藏代价：能力退化

Easy generation can reduce difficult practice - 轻易生成答案，会减少困难练习。

Students increasingly - 学生越来越：

- read / write less code
  减少代码阅读与编写
- reason less formally
  减少形式化推理

An observation in my CS courses lately - 我最近在计算机课程中的观察：

Pen-and-paper pseudocode performance has become dramatically uneven.
纸笔伪代码能力出现了极大的分化。

A potential suspect: AI convenience
一个可能原因：AI便利性

Remove friction and you may *remove learning*.
消除摩擦，也可能消除真正的学 习。

Remedial suggestions for students
对学生的建议

AI as ***partner***, not substitute - AI应作为伙伴，而非替代品

- Always start alone, ask for hints if stuck – 先独立思考，遇到困难时再寻求提示
- Challenge outputs – 主动质疑AI输出结果
- Request explanations of tradeoffs – 要求解释不同方案之间的权衡
- Compare multiple approaches – 比较多种解决方案

Learn how to think first. Then look for support.
先学会思考，再寻求辅助。

Remedial suggestions for teachers
对教师的建议

Teaching in an AI world: ***emphasize reasoning over output***
AI时代的教学：***重视推理，而非结果***

Include code reading, oral explanations, pen-and-paper exercises
加入代码阅读、口头解释、纸笔练习

Redesign assessments
重新设计考核方式

Measure ***understanding***, not the students' output.
评估理解，而不是只评估学生产出的结果。

Classroom coexistence with AIs
课堂中的人机共存

AI agents - AI智能体

- ~~Magical tutors 神奇的导师~~
- ~~Academic threats 学术威胁~~
- Powerful support tools 强大的辅助工具
- Persuasive language weavers (*sweet talkers*) 擅长“甜言蜜语”的语言织造者
- ***Absolutely*** incapable of independent thought 完全不具备真正独立思考能力

The challenge is to ensure everyone understands the *limitations of AI*.
真正的挑战，是确保每个人都理解AI的局限性。